The experiment checklist

The chain of every session: split, baseline, rule, model, errors, with the checks that keep each step honest.

1 Read the table

Session 1

- What is one row? Which column is the target? Which column is an ID that should not be a feature?
- Count duplicates, missing values and impossible values. Look at the share of each outcome.
- Plot each feature against the outcome before modelling.

2 Clean, then split

Sessions 1, 2, 4

- Clean what step 1 found (duplicates, impossible values) BEFORE splitting.
- Split into training, validation and test; stratified; fix a seed. Put the test set away.
- Split by the unit that must not be shared (by file or by person, not by row).

3 Baseline and a hand-written rule

Sessions 0, 4

- Score always answering the most common class. On imbalanced data add recall or precision.
- Write the rule you would have written without ML. That is the bar a model has to clear.

4 Preprocess inside a pipeline

Session 2

- Every learned number (median, scale, categories) comes from the training rows only.
- Use a Pipeline and cross-validation: report the mean and the spread, not one split.

5 Compare models on one protocol

Session 3

- Same split, same metric. Scale features for k-NN and logistic regression.
- Compare training and test accuracy: a big gap is overfitting. A point or two does not rank models.

6 Choose the decision with a written policy

Session 4

- Decide which mistake costs more BEFORE looking at a plot.
- Tune hyperparameters with cross-validation on the training rows; choose the threshold on the validation set.

7 Lock, then open the test set once

Sessions 4, 9

- Fix model, hyperparameters and threshold. Score the test set one time and report it as it is.
- Put an interval on it: a score on a few hundred cases can move by several points.

8 Read the errors

Session 0

- Look at the failures. Is the limit the data, the labels or the model?

9 Suspect results that are too good

Session 2

- A perfect score is a warning: look for a column that only exists after the outcome, or a step fitted before the split.

10 Neural networks and images

Sessions 7 to 9

- Train on 10 examples first: a healthy model gets them all right. Read training and validation loss together.
- Change one thing between runs; turn dropout off to evaluate; with few images, start from a pretrained backbone.

Source: the notebooks of the AI Journey basic course (third edition). Every number the course quotes is printed by its notebooks.